StyCLIP: StyleTransfer in Few-Shot Learning

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Introduction & Motivation

- **Context:** Rapid digitization of art collections demands automated artist classification systems for large-scale analysis.
- **Challenge:** Few-shot visual artist classification – identifying artist by their works with limited labeled examples per artist.
- **Two Main Goals:**
  1. **Comparative Analysis:** Test different CLIP-based strategies to identify the best approach for few-shot artist classification.
  2. **Style Enhancement Hypothesis:** Investigate whether explicitly integrating style representations (Gram matrices) into CLIP improves artist-specific pattern recognition.
- **Contribution:** StyCLIP – a CLIP extension incorporating dedicated style-aware features, evaluated against standard CLIP-based methods.

Related Work

Style Transfer

- Gatys et al. [3]: Deep networks encode style in intermediate layers; Gram matrices capture style.

Few-Shot Learning (FSL) & Meta-Learning

- Prototypical Networks [6]: Learn to learn from small samples.

Contrastive Language-Image Pretraining (CLIP)

- CLIP [5]: Shared embedding space for images and text.
- TIP-Adapter [8]: Non-parametric cache model for FSL.
- APE [9]: Adaptive prior refinement of CLIP features.
- GDA-CLIP [7]: Gaussian-based adaptation.
- TIMO [4]: Mutual guidance between text/image modalities.

Dataset

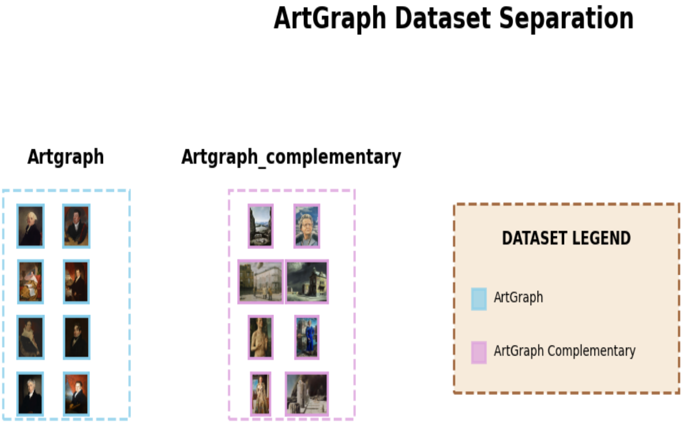


Figure: ArtGraph Datasets Overview

Episode Example for FSL

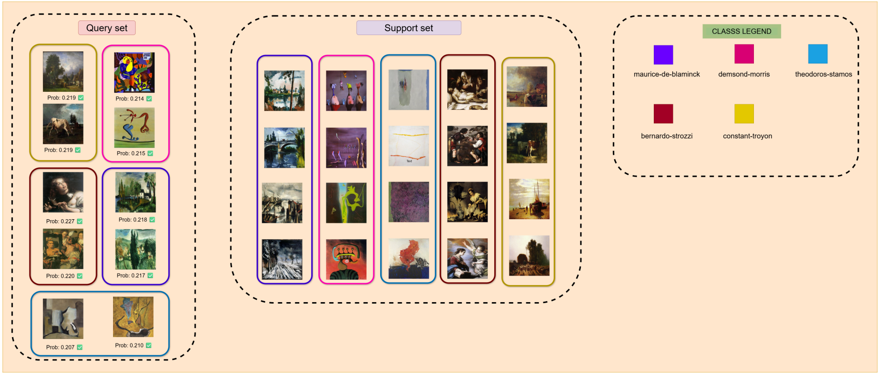
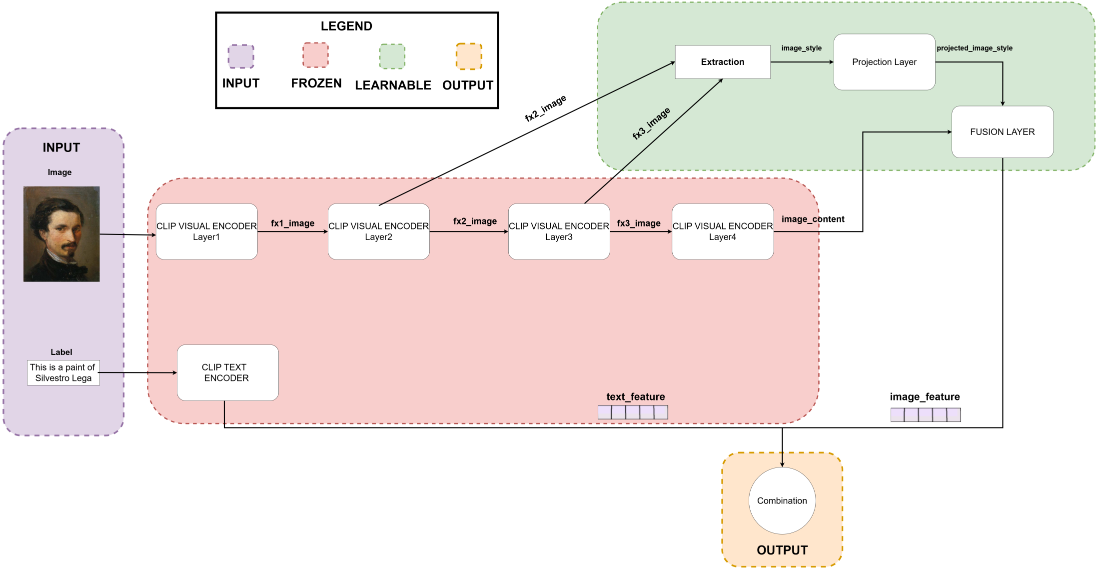


Figure: Example episode: 5-Way, 4-Shots ,10 query examples

StyCLIP: Architecture Overview



StyCLIP: Key Components

- **Three-Stage Architecture:**
  1. **Gram Matrix Extraction:** Extract style correlations from ResNet-50 intermediate layers

     $$G_{i,j}^{(l)} = \frac{1}{H \times W} \sum_{k=1}^{H \times W} F_{i,k}^{(l)} \cdot F_{k,j}^{(l)}$$

     - $F^{(l)} \in \mathbb{R}^{C \times H \times W}$: feature maps from layer l (reshaped to $C \times HW$)
     - $G_{i,j}^{(l)}$: correlation between channels i and j (captures style patterns)
     - Normalization by $H \times W$ ensures scale-invariance across image sizes
  2. **Style Projection:** Compress high-dimensional Gram matrices into compact representations

     $$\mathbf{s} = \mathrm{MLP}(\mathrm{concat}[\mathrm{vec}(\mathrm{triu}(G^{(2)})), \mathrm{vec}(\mathrm{triu}(G^{(3)}))])$$

  3. **Adaptive Fusion:** Integrate semantic and style features via bottleneck adapter

     $$\mathbf{f}_{\text{fused}} = \mathrm{Adapter}([\mathbf{F}_{\text{v}}; \mathbf{s}])$$

  - $[\mathbf{F}_{\text{v}}; \mathbf{s}]$: concatenation of CLIP features (512d) and style features (256d)
  - Adapter: bottleneck MLP ($768 \rightarrow (\textit{hyperParamDim}) \rightarrow 512$ dimensions)

StyCLIP: Training Strategy

- **Frozen Backbone:** Original CLIP (RN50) visual and text encoders are frozen.
- **Trainable Components:** Only the new style projection layers and fusion adapter are optimized.
- **Rationale:**
  - Preserve pre-trained semantic knowledge from CLIP.
  - Reduce computational cost.
  - Enable efficient fine-tuning on artistic datasets.

Hyperparameter Tuning

- Optuna [1] used for:
  - Fusion bottleneck dimension {64, 128, 256}
  - Dropout rate [0.0, 0.5]
  - Learning rate log-uniform $[10^{-5}, 10^{-3}]$
  - Weight decay log-uniform $[10^{-6}, 10^{-3}]$

Experimental Setup

Meta-Learning Training for StyCLIP

- Prototypical network approach.
- Episodic training: 10-way (N=10 artists), K-shot (K=1, 4) classification, 3-query (Q=3)

Evaluation Protocol

- **Method:** Using every CLIP-based method (e.g., TIMO, TIP-Adapter) with every backbone.
- N-way classification (all available artists in the test subset).
- Tested several times across K $\in \{1, 2, 4, 8, 16\}$ shots for support set.
- **Metrics:** Accuracy, Macro-Precision, Macro-Recall, Macro-F1.

Goal 1: Comparative Analysis of CLIP-based Methods

Best performing combinations for few-shot artist classification:

- **TIMO and TIMO-S** emerge as the most effective few-shot learning methods

Top Performers: 16-Shot Classification Accuracy (%)

Method	Backbone	Accuracy (%)
TIMO-S	**RN50**	**73.46**
TIMO	**RN50**	**73.36**
GDA-CLIP	RN50	73.06
APE	RN50	62.17
TIP-Adapter	RN50	52.91

Key Finding: TIMO-based methods with RN50 backbone achieve optimal performance for few-shot artist classification.

Goal 2: Style Enhancement Hypothesis - Results

Hypothesis: Explicit style representations (Gram matrices) improve artist classification.
Result: Hypothesis **rejected** - StyCLIP consistently underperforms standard CLIP variants.

StyCLIP vs Standard CLIP: 16-Shot Accuracy (%)

Method	RN50	StyCLIP_FSL1	StyCLIP_FSL4
TIMO-S	**73.46**	39.20	48.75
TIMO	**73.36**	37.00	43.27
GDA-CLIP	**73.06**	41.19	53.36
APE	**62.17**	6.73	7.98

- **Performance Gap:** Standard RN50 outperforms best StyCLIP variant by ∼25 percentage points
- **Statistical Significance:** T-tests confirm significant differences ($p < 0.05$)
- **Consistency:** Pattern holds across all shot configurations (1, 2, 4, 8, 16)

Performance Analysis Across Shot Configurations

Shot Config	TIMO-S + RN50	TIMO-S + StyCLIP_FSL4
1-shot	45.96%	7.31%
2-shot	48.65%	15.38%
4-shot	57.28%	26.36%
8-shot	65.08%	40.89%
16-shot	73.46%	48.75%

- Both approaches improve with more support examples
- **Performance gap persists** across all configurations

Conclusion & Discussion

- **Main Finding:** Integrating explicit style-aware features via Gram matrices (StyCLIP) did **not improve** few-shot artist classification performance.
- **Implication:** Explicit Gram matrix-based style modeling might interfere with CLIP's pre-trained semantic representations rather than enhance them for this task.
- **Future Directions:** Alternative approaches like attention-based style integration or domain-specific fine-tuning (without architectural changes) might be more fruitful.

Acknowledgments and Code

- This work was proposed by PhD student **Nicola Fanelli** and **Prof. Giovanna Castellano** at the University of Bari

- The code for this project is available on GitHub:

  `https://github.com/Fonty02/ComputerVision/tree/main/TIMO`

Thank You! 🚀

Appendix: Hyperparameter Optimization Highlights

StyCLIP FSL1 (1-shot trained)

- Bottleneck: 64
- LR: 6.01×10^{-5}
- Dropout: 0.0
- Weight Decay: 4.83×10^{-4}
- Valid Acc: 57.07%

StyCLIP FSL4 (4-shot trained)

- Bottleneck: 128
- LR: 1.02×10^{-5}
- Dropout: 0.1
- Weight Decay: 1.03×10^{-6}
- Valid Acc: 60.53%

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